



A Virtual Reality Intervention to Reduce Schizophrenia-Related Stigma: Qualitative Findings on Mechanisms of Change, Perceived Advantages, and Implementation Challenges

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Abstract

Immersive technologies are gaining attention as tools to improve public understanding of schizophrenia-related stigma; yet little is known about how users make sense of these interventions, or what mechanisms underpin their impact. Inclúyete-VR, a virtual reality program designed to promote empathy and reduce stigma toward people with schizophrenia, has previously shown significant effects in a randomized controlled trial. This qualitative study explored participants' perceptions of mechanisms of change, perceived benefits, and challenges following a virtual reality intervention implemented as part of a randomized controlled trial. Semi-structured interviews were conducted with 62 university students from the experimental arm of the parent RCT, and data were analyzed using directed qualitative content analysis. Participants described three interconnected processes that helped reduce stigma: (1) gaining a more accurate and concrete understanding of schizophrenia; (2) experiencing immersive perspective-taking that fostered embodied empathy; and (3) re-evaluating previously held stereotypes and social distance. Users valued the program's intuitive interaction, engaging and realistic environments, and recovery-oriented scenarios, which offered a hopeful and nuanced view of schizophrenia. At the same time, they identified areas for improvement, including clearer in-game guidance, enhanced sensory and graphic quality, cultural adaptation of specific scenes, and more intuitive navigation controls. Overall, Inclúyete-VR was perceived as an accessible and meaningful anti-stigma tool that supports experiential learning. These findings highlight the potential of VR to complement traditional educational and clinical strategies, particularly in training contexts, where fostering empathy, reducing fear-based assumptions, and promoting recovery-oriented practice remain central challenges.

Keywords Stigma · Virtual reality · Serious games · Qualitative · Schizophrenia · Psychosis

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Background

Stigma toward people with mental illness is a major global public health concern, contributing to social exclusion, increased suicide risk, and persistent barriers to help-seeking and social participation, particularly among people with severe mental illness such as schizophrenia (Corrigan et al., 2014; Kågstöm et al., 2025; Kruse & Dodell-Feder, 2025). Despite growing awareness, individuals with psychotic disorders continue to be portrayed as dangerous or unpredictable, reinforcing fear and avoidance rather than inclusion and recovery-oriented care (Dubreucq et al., 2021). This study focuses on public stigma, defined as stereotypes, prejudice, and discriminatory reactions held by the general population toward people with mental illness (Corrigan et al., 2012), which shapes population-level outcomes, influences the quality of care, and has relevant policy implications (Ahad et al., 2023).

Over recent decades, anti-stigma strategies such as public education and campaigns, curriculum-based initiatives and contact with people with lived experience have shown consistent benefits (Corrigan et al., 2012; Wong et al., 2024), but their impact is often limited by implementation challenges, variability in fidelity and contact quality, and inconsistent replication across settings (Kemp et al., 2019; Rodríguez-Rivas, 2022). These limitations are particularly salient among young people and students, for whom traditional approaches may be less engaging or insufficiently connected to real-world experiences of mental illness and recovery (Thornicroft et al., 2016; Rodríguez-Rivas, 2021). Stigma also persists among clinical trainees and practitioners, affecting therapeutic engagement and confidence in working with psychosis (Bielenberg et al., 2021; Wang et al., 2025; Nyblade et al., 2019), highlighting the need for educational strategies that foster empathy, accurate understanding, and reflective practice (Antoine et al., 2023).

In this context, technology-based interventions have gained attention as complementary tools to address public stigma (Johnson et al., 2022). A meta-analysis identified nine randomized trials using digital strategies, including video games, virtual reality (VR), and augmented reality, with consistent medium effects on reducing public stigma (Rodríguez-Rivas et al., 2022). While these findings support the potential of digital approaches, they also reveal gaps related to mechanisms of change, long-term effects, and cultural and educational-context adaptation.

Among digital interventions, immersive VR has emerged as a promising modality, enhancing cognitive and emotional engagement and promoting perspective-taking and empathy through embodied learning (Barreda-Ángeles et al., 2020; Yuen & Mak, 2021). However, VR-based stigma

interventions do not produce uniform effects: symptom-only simulations may reinforce fear when not embedded within a recovery-oriented framework (Brown et al., 2010), whereas programs integrating symptom experiences with psychoeducation, hopeful narratives, and indirect contact show more consistent outcomes (Waqas et al., 2020), underscoring the need to clarify active components and guide cultural adaptation (Daniel et al., 2024).

Inclúyete-VR is one of several VR-based programs designed to reduce stigma toward serious mental illness (e.g., Lem et al., 2024; Tay et al., 2025; Yuen & Mak, 2021). Its effectiveness has been demonstrated in a randomized controlled trial with university students, showing significant short-term reductions in global stigma and key dimensions such as perceived dangerousness, avoidance, and lack of solidarity compared with a neutral VR control (Rodríguez-Rivas et al., 2024), consistent with broader evidence on immersive technology-based interventions (Tay et al., 2025; Yuen & Mak, 2021).

However, quantitative trials alone cannot explain how users experience these interventions, which components are most meaningful, or which psychological and relational processes drive change (Dahler-Larsen, 2023; Wasti et al., 2022). Most VR stigma studies have focused on feasibility, usability, and attitudinal change, with fewer investigations examining subjective experiences, emotional and cognitive responses, and implementation challenges in real-world settings, including the role of contextual factors such as cultural relevance, navigation, and sensory design in facilitating or hindering learning and empathy (Syed, 2025). This gap is particularly relevant for Inclúyete-VR. Although the parent RCT demonstrated significant short-term reductions in global stigma and in three dimensions (dangerousness-fear, lack of solidarity, and avoidance), the quantitative findings did not clarify how participants interpreted the immersive experience, which components they perceived as most meaningful, or how the sequence from psychotic-like simulation to recovery-oriented scenarios may have contributed to those changes.

Understanding users' experiential perspectives is therefore essential to clarify how VR stigma-reduction tools produce change, identify design elements that facilitate or hinder engagement, and guide cultural adaptation, acceptability, and scalability in real-world contexts.

Objective

To address these gaps, the aim of the present study was to explain, from the users' perspective, how participants experienced Inclúyete-VR and how these experiences may help

interpret the stigma-reduction effects observed in the parent randomized controlled trial. Specifically, we explored: (1) the mechanisms and changes associated with stigma reduction as described by participants after the VR experience; (2) the possible advantages of *Inclúyete*-VR that enhanced engagement and learning; and (3) the challenges and potential improvements they identified regarding the software's design and implementation.

Methods

Study Design and Participants

We conducted a qualitative explanatory phase study (Hwang et al., 2025) embedded within a randomized controlled trial evaluating *Inclúyete*-VR. The parent trial was registered at ClinicalTrials.gov (NCT05393596; May 23, (Daniel, et al., 2022)).

Participants were undergraduate and graduate students from a private university in Chile who had taken part in the experimental arm of the trial. Recruitment followed convenience sampling by trained research staff in shared campus spaces. Inclusion criteria were: ≥ 18 years, current university enrollment, and written informed consent.

The final sample comprised 62 students aged 18–35 years ($M=21.8$, $SD=2.9$), 59.7% women, representing health sciences (29%), psychology (29%), engineering (23%), and other fields (19%). All experimental-group participants were invited to the qualitative phase. All 62 participants allocated to the experimental arm completed the post-intervention interview, resulting in a 100% uptake rate. Interviews were conducted immediately after the VR session during the same study visit. No delayed qualitative interviews were conducted.

The study was approved by the Ethical Scientific Committee of XX (ID 2022-43). Participation was voluntary and confidential, without compensation. Data were encrypted, de-identified, and accessible only to the principal investigator.

Virtual Reality Intervention (*Inclúyete*-VR)

Inclúyete-VR is an immersive program developed in Spain through a co-design process with mental health service users and professionals (Cangas & Galván, 2020). It aims to reduce stigma toward schizophrenia by combining simulations of psychotic-like experiences with recovery-oriented narratives and psychosocial resources, approximating subjective aspects of psychosis within a framework of recovery and social inclusion.

The experience begins with auditory and visual disturbances and progresses to a virtual “crisis center” with six interactive rooms representing psychosocial recovery domains (e.g., arts, sports, occupational and therapeutic activities, media, and daily living tasks). The sequence shifts from distressing to hopeful, recovery-oriented experiences.

The intervention was delivered individually using Oculus Quest 2[®] headsets in a university-based immersive technology environment. Participants received brief training in VR navigation, and sessions were monitored by research staff. The experience lasted approximately 20–30 min, with written instructions on safety and navigation.

Interview Procedure

Immediately after the intervention, participants completed semi-structured individual interviews conducted on site by trained research assistants. Two undergraduate psychology students conducted the interviews after standardized training in qualitative interviewing and ethical procedures. Interviews lasted 10–15 min, were conducted in Spanish, audio-recorded, transcribed verbatim, anonymized, and translated into English for publication.

The interview guide was informed by public stigma dimensions, prior VR stigma research, and outcomes of the parent RCT. It addressed three areas aligned with study aims: perceived mechanisms, benefits, and challenges, using open-ended questions on inclusion, strengths, and areas for improvement. Specifically, participants were asked: (1) “Do you think this software promotes social inclusion of people with mental health problems? How?”; (2) “What strengths or positive aspects stand out from your experience with the VR intervention?”; and (3) “What weaknesses or areas for improvement would you suggest for future versions of the software?”

Data Analysis

Data were analyzed using directed qualitative content analysis (Elo & Kyngäs, 2008; Hsieh & Shannon, 2005). Three a priori domains—perceived mechanisms of change, perceived benefits, and perceived challenges—guided initial data organization. These domains were derived from the study aims, the structure of the interview guide, prior stigma-reduction theory, and the outcomes assessed in the parent RCT. Because this qualitative component was embedded within a randomized controlled trial, the interview guide was defined a priori and not iteratively modified, with the aim of capturing the range and meaning of experiences within the intervention group.

Transcripts were read repeatedly for immersion. Meaning units relevant to the study aims were identified and

coded inductively within each domain, so that participants' accounts were not forced into predefined themes. Two members of the research team independently reviewed the transcripts and developed initial codes. Codes were then compared, discussed, and refined through team meetings. Disagreements were resolved by consensus, and codes with similar meanings were grouped into descriptive themes.

The analysis focused on describing and organizing participants' experiences rather than generating latent or interpretive themes. Analytic rigor was supported through iterative team discussions, comparison of coded excerpts with original transcripts, and ongoing refinement of themes to ensure coherence and consistency.

Given the explanatory and directed nature of the qualitative component, we assessed informational sufficiency rather than theoretical saturation. Recruitment was determined by the parent RCT sample, and the final set of interviews provided recurrent and coherent accounts across the three domains.

Research Team and Reflexivity

Interviewers were not involved in the development of the VR software. The research team had expertise in

mental health stigma and technology-based interventions and participated in the parent RCT. Potential social desirability bias was addressed by emphasizing voluntariness and encouraging critical feedback. Coding consistency was supported through independent transcript review, comparison of coded excerpts, consensus discussions, and iterative refinement of the thematic structure. Participant validation was not conducted because interviews were brief, immediate post-intervention accounts embedded within the RCT protocol.

Findings/Results

To facilitate understanding of the findings, this section is organized into three overarching domains. Domains 1 and 2 include descriptive themes, whereas Domain 3 presents a synthesized account of challenges and improvement opportunities. For each domain, we present a brief interpretive description followed by selected illustrative quotations that represent the strongest and clearest expressions of participants' experiences. Figure 1 presents a conceptual map of the qualitative findings. Table 1 provides an overview of the domains, themes, and representative quotations.

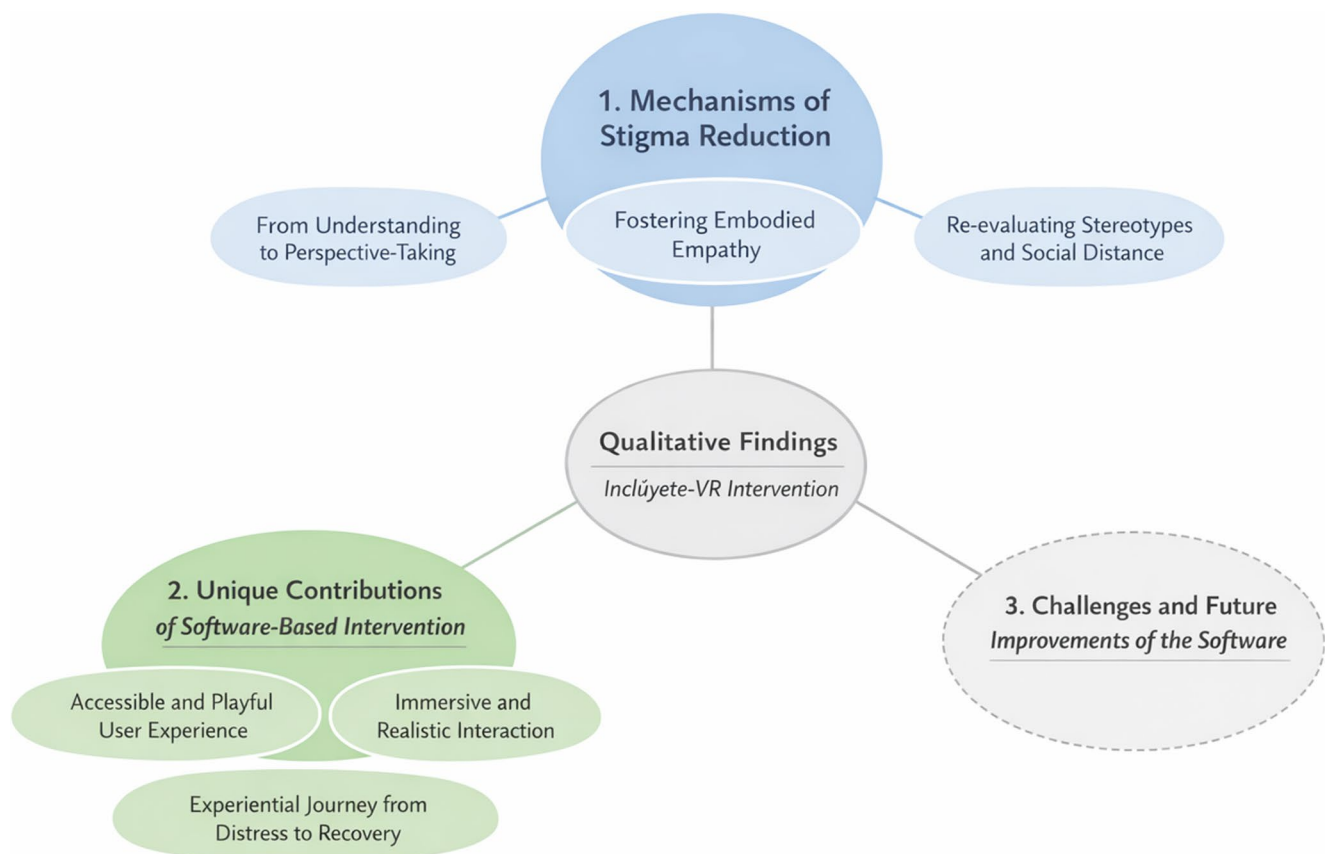


Fig. 1 Conceptual map of qualitative domains and themes identified in the Inclúyete-VR intervention

Table 1 Overview of domains, themes, and representative quotations

Domain	Theme	Representative quotation
Mechanisms of Stigma Reduction	From Understanding to Perspective-Taking	<i>"A lot of fear comes from not knowing, and understanding what the realities are gives you more peace of mind."</i> (E-41) <i>"It gives you a kind of approach to what living something like this would be like. Once you've had that experience, I feel you develop more empathy, and that relates to inclusion, because discrimination often comes from ignorance. So learning is like taking steps toward the solution, I think."</i> (E-14)
	Fostering Embodied Empathy	<i>"Truly putting yourself in someone else's place means living it in first person through the game, which gives you a much more real impression of how they live, beyond just words."</i> (E-47)
	Re-evaluating Stereotypes and Social Distance	<i>"It takes away the stigma that they hear voices all the time. That does happen, but not constantly. They're not 'crazy' like movies portray. It helps you see they're just like other mentally healthy people."</i> (E-20) <i>"Before playing this, schizophrenia was something I preferred to stay away from. After this, I can accept it a bit more. If a person with schizophrenia wants to work, they're immediately judged because of their condition. This can help reduce that judgment in work settings or relationships."</i> (E-26)
Unique Contributions of Software-Based Intervention	Accessible and Playful User Experience	<i>"I found it interesting that in the little rooms you can see the activities that people with this condition can do. I thought it was interesting that I myself had fun doing the activities—you need concentration and motor skills to complete them."</i> (E-28) <i>"The biggest strength for me is that it's a fun system. We're not used to this type of video or these kinds of realities, and it allows people to learn in a more enjoyable way."</i> (E-6)
	Immersive and Realistic Interaction	<i>"When the monsters appeared, I didn't expect them to come so fast or get so close. I felt it gave me a sense of what a person with schizophrenia might experience."</i> (E-21)
	Experiential Journey from Distress to Recovery	<i>"The first experience, the ghost one, is very interesting. It's quite a strong impact to start with, so I really liked it because it reflects a reality that is often unknown. It's not common, and it's very striking. So, I liked that it started that way and then immediately showed how it is managed, and later how it feels. I really liked the order."</i> (E-24)
Challenges and Future Improvements of the Software		<i>"It allows companies and people to experience those adapted scenarios... to create environments where individuals can function comfortably and to empathize. It could have a big impact on society."</i> (E-25) <i>"Maybe during the rooms, it could feel more like a guided tour. For example: 'this room has these purposes.' I felt a bit lost, and without someone telling me what to do, I wouldn't have known. A more guided format would help."</i> (E-28) <i>"Graphics and design could be improved, although that might not fully align with the program's goals."</i> (E-23) <i>"Maybe the space could be larger so you're not constantly thinking you're going to bump into something."</i> (E-42)

Domain 1: Mechanisms of Stigma Reduction

Because the parent RCT showed significant reductions in global stigma, dangerousness-fear, lack of solidarity, and avoidance, this domain was interpreted as participants' experiential account of the processes that may have contributed to those quantitative changes. This domain integrates the subjective processes participants identified as central to understanding how the intervention influenced their perceptions of people with schizophrenia. Their accounts

reveal a sequence of experiential change: first, the immersive experience fostered a more accurate understanding of the condition; second, this understanding evolved into empathy and emotional awareness; finally, these changes led to adjustments in perspectives and attitudes, reducing stereotypes and promoting more inclusive views. Together, these themes reflect a process of perspective-taking and embodied empathy contributing to attitudinal change.

Below, we present the three themes that compose this domain, along with illustrative quotations.

From Understanding to Perspective-Taking

This theme reflects how the intervention supported both informational and experiential approaches to schizophrenia. Participants reported gaining meaningful insight into symptoms, everyday experiences, treatment contexts, and therapeutic options. This understanding reduced fear, corrected misconceptions, and fostered a more humanized and realistic view of schizophrenia. In turn, this grounded comprehension facilitated empathy, recognition of symptom variability, and appreciation of recovery-oriented environments in practice. Illustrative quotes include:

“I think it helps you understand people in these situations, like their therapies or parts of their world. With the software, we can see the therapy and face it as a real situation.” (E-11).

Fostering Embodied Empathy

This theme groups accounts highlighting the software’s capacity to generate emotionally meaningful experiences. Participants described that “putting oneself in the other’s shoes” became a concrete embodied experience within the immersive environment. Empathy emerged from experiencing sensations associated with psychotic symptoms, witnessing exclusion, and visualizing emotional burden, while also encountering contexts of support and recovery. Illustrative quotes include:

“I think this can really help people put themselves in someone else’s place before judging them and truly understand what it feels like to be in that situation... it really helps us put ourselves in the shoes of people who have this condition.” (E-1).

Re-Evaluating Stereotypes and Social Distance

This theme captures how understanding and empathy led participants to re-evaluate stereotypes and feel more willing to engage socially with people with schizophrenia. Participants reported that the intervention reduced unjustified fears, challenged stereotypes of dangerousness, and promoted a more inclusive view regarding work, daily life, and interaction with people with schizophrenia. This change emerged as a shift in how they saw, interpreted, and valued affected individuals.

Participants described moving away from avoidance or apprehension and toward a more humanized understanding of the condition, recognizing that people with schizophrenia navigate their symptoms in diverse and often non-threatening ways. Illustrative quotes include:

“Before playing this, schizophrenia was something I preferred to stay away from. After this, I can accept it a bit more... This can help reduce judgment in work settings or relationships.” (E-26).

Domain 2: Unique Contributions of Software-Based Intervention

This domain encompasses design and user experience features identified as core strengths of the software. Participants generally perceived Inclúyete-VR as accessible, engaging, and emotionally appealing, with a design facilitating immersion and progressive content delivery. These characteristics supported usability, emotional involvement, and psychoeducational impact. Rather than technical attributes, these elements were experienced as active contributors to the intervention’s effects. Three themes were identified.

Accessible and Playful User Experience

Participants consistently described *Inclúyete-VR* as intuitive, easy to operate, and supported by a coherent activity design. This usability helped them stay focused on the emotional and experiential aspects of the program rather than on navigating controls or understanding mechanics. Users noted that the clarity of the tasks reduced cognitive load, allowing immersion and engagement to occur more naturally. The intervention’s accessible design appeared especially beneficial for those with limited exposure to VR technologies, who still reported feeling comfortable and “guided” throughout the experience.

“It’s a game that is very easy to use and understand; people can handle it easily after the explanation is given.” (E-15).

Beyond usability, participants highlighted a strong playful component that made the experience feel enjoyable rather than effortful or clinical. This sense of “being in a game” functioned as a motivational driver: it reduced initial resistance, facilitated emotional openness, and sustained attention across different scenarios. The pleasant and playful tone of the activities created a balance between learning and enjoyment, helping users engage with difficult content without feeling overwhelmed.

“You feel like you’re playing; it doesn’t feel like a treatment or something that’s supposed to ‘fix’ you. It just feels like playing, like happiness.” (E-2).

Immersive and Realistic Interaction

Most participants highlighted the sense of presence, sound quality, physical interaction with objects, and emotional realism of difficult situations. This immersion enabled “living” the experience from the inside, reinforcing the program’s educational impact. Illustrative quote include:

“I found it incredible, really good, extremely real. I thought it was awesome that you could move around and grab things. I had done 4D or 3D seats before, but I had never actually been able to grab things. I found it completely new.” (E-8).

Experiential Journey from Distress to Recovery

Participants valued the structure that moved from an emotionally intense opening scenario to activities associated with calmness, agency, and wellbeing. This sequencing helped them understand the contrast between symptom-related distress and recovery possibilities, allowing the transition from crisis to recovery to be experienced as a coherent emotional journey rather than disconnected scenes.

Participants reported that starting with a difficult moment generated impact and awareness, while ending with familiar and enjoyable activities fostered a more balanced and humane view of schizophrenia. This progression softened the emotional tone of the experience and conveyed a hopeful message, countering stereotypical “all-or-nothing” representations and highlighting that positive moments are possible and meaningful. Illustrative quotes include:

“What I liked the most is that it started with the difficult part and ended with the more enjoyable activities, like shooting at little figures, like in the games everyone plays. It makes you go through the whole reality, but at the end it leaves you with a pleasant feeling — like we painted, watered plants, spent time with a dog, played basketball. I feel that it leaves you with something, with a happier and hopeful feeling. It shows that you can do things; it’s not like a person with schizophrenia is like that all the time. It’s not all-or-nothing — there can also be good moments.” (E-12).

Domain 3: Challenges and Future Improvements of the Software

This domain summarizes participants’ observations regarding areas for improvement in future versions of the program. Although the intervention was highly valued, participants identified limitations related to educational reach, cultural adaptation, and technical and navigational features.

Participants highlighted the program’s potential to educate the public about schizophrenia but noted its limited current reach. They suggested expanding implementation to community settings and increasing visibility to broaden access to VR-based educational experiences, positioning VR as a bridge between clinical knowledge and public understanding. Overall, visibility and accessibility were identified as key factors for transforming Inclúyete-VR into a widely available educational resource. Illustrative quotes include:

“I would like it to be applied in a more widespread way to psychoeducate people...” (E-15).

Participants also emphasized the need for cultural adaptation. Some scenarios felt disconnected from everyday Chilean contexts, which weakened identification and emotional engagement. They suggested incorporating more locally relevant environments, activities, and objects to enhance contextual relevance.

“It would be good to explain the radio room beforehand and adapt it more to Chile...” (E-15).

Another recurring limitation concerned the lack of guidance during the experience. This finding does not contradict participants’ reports that the software was generally intuitive and easy to operate. Rather, it suggests a distinction between basic usability, understood as the ability to navigate and interact with the software after initial explanation, and pedagogical scaffolding, understood as the need for clearer in-game guidance to understand the purpose and meaning of each activity.

“Having some prior instruction on how the system and the controls work would help...” (E-13).

Participants further proposed incorporating a guide figure or narrative cues to replace researcher explanations and support autonomous navigation. They also suggested voice-based instructions or alternative modalities to enhance accessibility for users with diverse literacy or cognitive profiles.

“For me, it would need someone guiding you...” (E-6).

From a technical perspective, participants identified visual and audio aspects that could be improved. Occasional glitches and limited graphical resolution disrupted immersion, while artificial or monotone voice recordings reduced emotional realism. These observations pointed to opportunities for technical refinement rather than dissatisfaction with the core design.

"The game's quality wasn't very HD, and sometimes it glitched." (E-9).

"The voice sounded generic and pre-recorded..." (E-18).

Finally, participants noted that the physical setup influenced immersion. Limited space generated discomfort and constrained movement, disrupting the flow of the experience and reducing embodiment. They suggested that a larger or better-defined physical area would enhance safety, fluidity, and realism. Illustrative quotes include:

"It really should be a more open space where I can walk..." (E-5).

Discussion

The aim of this qualitative study was to explain how participants experienced Inclúyete-VR within the context of an RCT and to clarify how their accounts may help interpret the stigma-reduction effects observed in the parent trial. Overall, the findings indicate that users interpreted the program through three interconnected processes targeting public stigma: gaining a more accurate understanding of schizophrenia, developing empathy through immersive perspective-taking, and revising previously held beliefs and attitudes. These processes were consistent with the quantitative reductions observed in global stigma, dangerousness-fear, lack of solidarity, and avoidance. These mechanisms align with theoretical expectations from stigma-reduction research while providing nuanced insights into the experiential processes through which VR-based interventions may operate in real-world contexts (Riva, 2022; Tay et al., 2025). Beyond individual attitudes, the findings suggest that VR may also influence expectations about recovery-oriented practice, highlighting modifiable components of system-level stigma.

More specifically, participants' accounts help interpret how the RCT effects may have occurred. Reports of a more concrete and humanized understanding of schizophrenia align with the reduction in global stigma, while descriptions of reduced fear and re-evaluation of stereotypes are consistent with the decrease in dangerousness-fear. Similarly, embodied empathy and perspective-taking may help explain reductions in avoidance and lack of solidarity, as participants described greater willingness to understand and support people with schizophrenia. In contrast, the parent RCT did not show a significant change in coercion, suggesting that beliefs about forced treatment or institutional control may require more targeted intervention components.

Participants described the VR experience as a concrete and experiential way of understanding schizophrenia, involving emotional engagement with daily life, symptom distress, and recovery practices. This supports evidence that public stigma often arises from misinformation and limited exposure (Antoine et al., 2023; Corrigan et al., 2012; Thornicroft et al., 2016) and underscores the potential of embodied learning to correct misconceptions more effectively than traditional didactic approaches (Tay et al., 2025). From participants' perspectives, VR may function as a knowledge-corrective tool that facilitates cognitive restructuring and challenges culturally reinforced stereotypes shaped by media and social narratives (Rodríguez-Rivas et al., 2022).

Empathy emerged as a second core mechanism, with participants identifying perspective-taking as a key pathway for reducing fear and avoidance (Yuen & Mak, 2021; Riva, 2022). Empathy was not only cognitive but also emotional and embodied, consistent with models of immersive learning in VR (Barreda-Ángeles et al., 2020). This process helps explain reductions in dangerousness-related fear and avoidance observed in the parent RCT and suggests a shift toward more positive relational attitudes toward people with mental illness.

The integration of simulated symptom experiences with recovery-oriented activities was central to these effects. Prior research shows that isolated symptom simulations may reinforce fear-based stereotypes (Brown et al., 2010), whereas interventions combining distress-related experiences with hopeful narratives and psychosocial resources yield more consistent outcomes (Waqas et al., 2020; Daniel et al., 2024). The sequential design from distress to wellbeing may therefore account for both the emotional impact reported by participants and the attitudinal improvements observed quantitatively.

Several design features strengthened learning and stigma-reduction processes. Participants highlighted usability and playfulness, suggesting that accessible interfaces and low cognitive load facilitate deeper engagement with emotionally sensitive content (Johnson et al., 2022; Rodríguez-Rivas et al., 2022). Immersion, driven by sensory realism and interaction, fostered a sense of presence associated with emotional resonance and cognitive change (Riva, 2022). Participants also valued the safety and calm of recovery-oriented environments, which are often underrepresented in clinical training focused primarily on symptoms or risk (Bielenberg et al., 2021; Nyblade et al., 2019). VR thus enabled exposure to psychosocial rehabilitation contexts that many participants had not previously encountered.

At the same time, participants identified areas for improvement, including navigation difficulties and the absence of onboarding guidance, consistent with challenges reported in other VR mental health interventions (Syed, 2025). Users also requested clearer in-game explanations and narrative framing, reinforcing evidence that guidance

and structure enhance the impact of VR anti-stigma programs (Tay et al., 2025; Rodríguez-Rivas et al., 2022).

Beyond design considerations, participants emphasized the need to expand the program's psychoeducational reach. Although Inclúyete-VR demonstrated effectiveness in controlled settings, its broader impact depends on accessibility and dissemination in community contexts, such as schools, workplaces, and public spaces (Wong et al., 2024; Corrigan et al., 2012). Future adaptations could also examine its use in professional and service-oriented contexts where stigma toward psychosis may affect interaction, decision-making, or support, including health professions education, emergency department personnel, first responders, educators, family members, and other groups who may interact with people experiencing psychosis. Finally, participants highlighted the importance of cultural adaptation, noting that some scenarios did not fully reflect Chilean everyday realities, a challenge widely documented in cross-context anti-stigma interventions (Daniel et al., 2024).

Overall, participants' accounts suggest that immersive and recovery-oriented VR may offer a distinctive contribution to anti-stigma efforts by integrating mechanisms difficult to achieve through traditional educational approaches. These findings position Inclúyete-VR within the growing ecosystem of technology-based anti-stigma interventions, where immersive tools may foster perspective-taking, emotional engagement, and knowledge correction beyond conventional methods (Rodríguez-Rivas et al., 2022; Lem et al., 2024; Tay et al., 2025).

Limitations

This study has several limitations. The sample included only university students from the experimental arm of an RCT, which may limit the transferability of the findings to other age groups, educational backgrounds, occupational contexts, and people with lower familiarity or receptiveness toward technology-based mental health interventions. The qualitative phase lacked a control group, restricting attribution of mechanisms to VR versus program content. Findings relied on brief, immediate self-reports, which may reflect initial impressions and social desirability. Because interviews were conducted immediately after the intervention and within the research context, participants' responses may have been influenced by the proximity of the experience and by the presence of research staff associated with the study. In addition, participant validation was not conducted, which may limit the confirmability of the interpretations. Future studies should include more diverse samples, delayed assessments, independent interviewers, participant feedback strategies, and mixed methods designs assessing knowledge and attitudes before and after the intervention.

Conclusions

From the users' perspective, *Inclúyete*-VR offers meaningful mechanisms for reducing stigma, particularly by improving understanding, fostering empathy and challenging stereotypes about schizophrenia. Participants valued its usability, immersive qualities and recovery-oriented structure, while also identifying clear opportunities for refinement, including cultural adaptation, improved guidance and enhanced sensory and technical design.

Overall, these findings reinforce the potential of VR as an accessible and engaging anti-stigma tool and point to specific developments needed to strengthen its effectiveness and applicability across educational, clinical and community settings.

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Author Contributions M.E. R-R.: Conceptualization; Methodology; Formal Analysis; Investigation; Data Curation; Project Administration; Supervision; Visualization; Writing– Review & Editing; P. V.: Methodology; Investigation; Formal Analysis; Writing – Review & Editing; A.J.C.: Conceptualization; Methodology; Writing – Review & Editing; A.M.: Validation; Writing – Review & Editing; L.C.: Validation; Writing – Review & Editing; D.F-O: Validation; Methodology; Writing – Introduction; Writing – Results; R.Q-S.: Validation; Methodology; Writing – Review & Editing; P.C.: Validation; Writing – Review & Editing.

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Data Availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval The study was approved by the Ethical Scientific Committee of the Facultad de Medicina—Clínica Alemana, Universidad del Desarrollo, Chile (ID 2022-43).

Consent to publish Not applicable.

Informed consent Informed consent was obtained from all individual participants included in the study.

Clinical trial number This qualitative study was conducted within a registered randomized controlled trial (ClinicalTrials.gov Identifier: NCT05393596), registered on May 23, 2022.

Author area of interest Mental health stigma; help-seeking behavior; virtual reality interventions; digital mental health; psychosocial interventions; adolescent mental health; qualitative and quantitative research.

Conflict of Interest The authors declare no conflict of interest.

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